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Programming Language, Spring 2024, Homework #1 (Due: 18:30, 03/05)

Problem 1: When converting a negative integer into 2's-complement binary notation, one must first represent the magnitude, then complement the bits, then add 1. Please have a proof the show that 2's-complement representation: $-x = 2^n - x$.

Problem 2:

a. What is the minimum number of bits that are required to uniquely represent the characters of English alphabet? (Consider upper case characters alone)
b. How many more characters can be uniquely represented without requiring additional bits?

Problem 3: Using 7 bits to represent each number, write the representations of 23 and -23 in signed magnitude and 2's complement integers.

Problem 4:

a. What is the largest positive number one can represent in a 12-bit 2's complement code? Write your result in binary and decimal.
b. What is the greatest magnitude negative number one can represent in a 12-bit 2'scomplement code? Write your result in binary and decimal.
c. What is the largest positive number one can represent in n-bit 2's complement code?
d. What is the greatest magnitude negative number one can represent in n-bit 2'scomplement code?

Problem 5: Using the smallest data size possible, either a byte, a halfword or a word, covert the following values into two's complement representation, write the value in hexadecimal:

a. -20
b. 114
c. -128
d. -129

Problem 6: What are the overflow and carry flags of the following operations? (Assume a four-bit system.)

	Carry	Overflow
1101 + 1100		
1101 - 1100		
1100 + 1010		
0100 - 0110		
0100 + 0010		
0100 + 0110		
1100 - 0110		

1. ***Homework will be available on E3.NYCU***
2. ***Return your homework with A4 sized paper. Different paper size will not be accepted.***
3. ***Note your name and ID number on the top of the paper.***
4. ***Write the report in the same sequence as the problems.***
5. ***Codes, results, figures and discussions should be added in the report.***
6. ***No cover required.***
7. ***No late submit.***